	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 4 – Organic Chemistry and Spectroscopic Analysis

SOLUTIONS / MARKING SCHEME

Date: Monday August 15th, 2022

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	42	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.
- You may keep the data book.

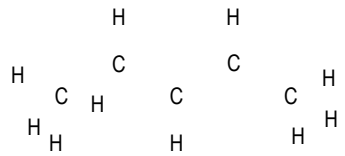
Electronic Devices

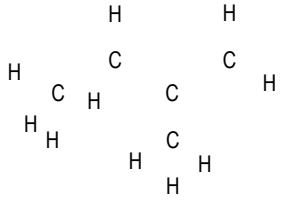
Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Q1.

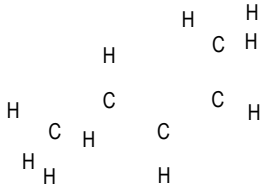
Consider the structure at the right.

Which of the structures shown below does ***not*** represent an *isomer* of this?

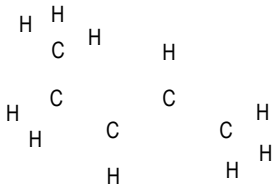




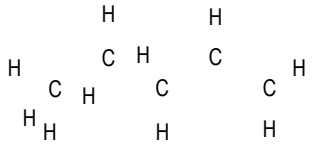
A



B



C



D

Q2.

Why is a trace of tetramethylsilane (TMS, $Si(CH_3)_4$) added to a sample before recording an NMR spectrum?

A It stabilises the sample to prevent chemical reactions occurring while in the spectrometer.

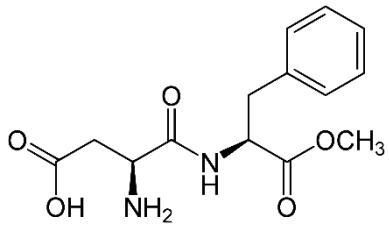
B It is an internal standard, producing a signal that is used as the basis for chemical shift measurement.

C It causes splitting of the signals, providing invaluable additional information about the structure.

D It allows the operator to check that the spectrometer is operating correctly.

Q3.

At the right is the structure of the artificial sweetener aspartame. Which of the following structural units are ***all present*** in the structure of this molecule?



A amino and ester group, and carboxy group

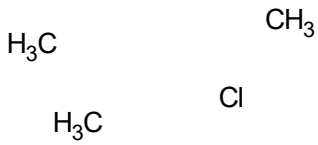
B ketone, carboxy, amino group

C amide and hydroxy group, and ester group

D carboxy, ketone, and amide groups

Q4.

The compound with the structure shown at the right would exhibit



A geometric (cis/trans) isomerism but not optical isomerism (chirality)

B optical isomerism (chirality) but not geometric (cis/trans) isomerism

C neither geometric (cis/trans) isomerism nor optical isomerism (chirality)

D both geometric (cis/trans) isomerism and optical isomerism (chirality)

Q5.

The balanced equation for hydroxylation of chlorethane using potassium hydroxide is

$CH_3CH_2Cl + KOH \rightarrow CH_3CH_2OH + KCl$

What effect would using LiOH instead of KOH have on the *atom economy* of this reaction?

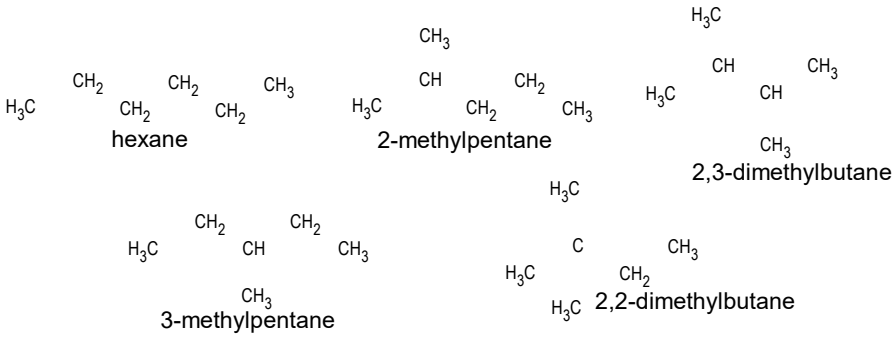
A The atom economy would be increased

B The atom economy would be decreased

C The atom economy would be unaffected

D Unknown. The effect on the atom economy must be determined experimentally

The five structural isomers of hexane shown here:



Q6. .

Of these, the ones that would show four signals in the ^{13}C NMR are

A 2-methylpentane and 3-methylpentane

B 3-methylpentane and 2,2-dimethylbutane

C 3-methylpentane and 2,3-dimethylbutane

D 2-methylpentane and hexane

Q7.

Butan-2-ol is oxidised using acidified potassium dichromate solution. Which of the following half equations correctly shows the oxidation of butan-2-ol?

A $C_4H_{10}O + H_2O \rightarrow C_4H_8O_2 + 4H^+ + 4e^-$

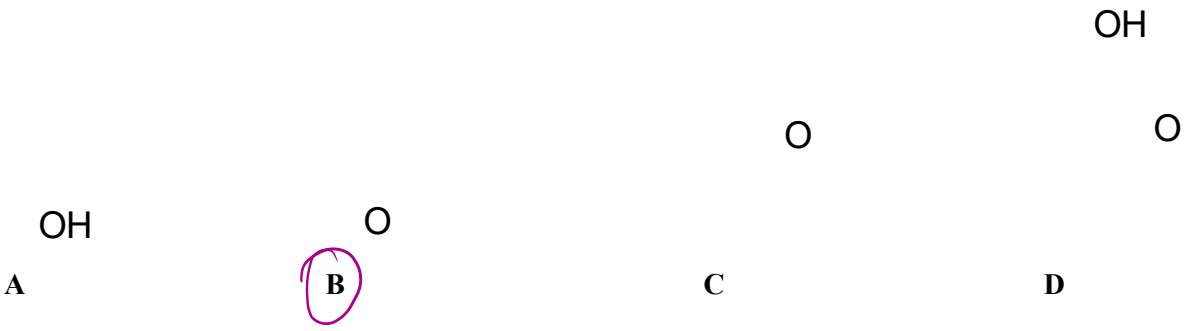
B $C_4H_{10}O \rightarrow C_4H_8O + 2H^+ + 2e^-$

C $C_4H_{10}O + H_2O \rightarrow C_4H_6O_2 + 6H^+ + 6e^-$

D $C_4H_{10}O \rightarrow C_4H_6O + 4H^+ + 4e^-$

Q8.

Which of the following molecules would produce a ^{13}C NMR spectrum including a signal with the ***highest*** chemical shift (δ) value?



Q9. **(4 marks)**

a) Which would have the higher boiling point, pentane (M=72.0 g mol^{-1}) or butan-1-ol (M=74.0 g mol^{-1})? Justify your prediction. (2 marks)

butan-1-ol (1) as it has stronger intermolecular bonding

pentane is capable of ***dispersion forces only*** (1)

butan-1-ol possesses an -OH group allowing for much stronger ***hydrogen bonding*** (1) in addition to dispersion forces

(note that the similar molecular masses/electron numbers suggests the dispersion forces would be similar in the two substances)

b) Which will be more soluble in water, hexanoic acid (M=116.0 g mol^{-1}) or octanoic acid (M=144.0 g mol^{-1})? Justify your prediction. (2 marks)

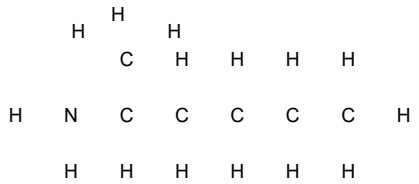
hexanoic acid (1)

- both possess a ***polar***, ***hydrophilic*** -COOH group allowing for comparable hydrogen bonding to water molecules
- hexanoic acid possesses a ***smaller, non-polar***, ***hydrophobic***, alkyl 'tail' so will be more soluble overall

(not all key terms necessary)

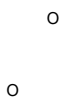
Q10. **(4 marks)**

a) Give a systematic name for each of the structures shown below. (3 marks)



hexan-2-amine

$CHC(CH_2)_3CH(CH_3)_2$ 6-methylhept-1-yne



ethyl octanoate

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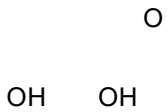
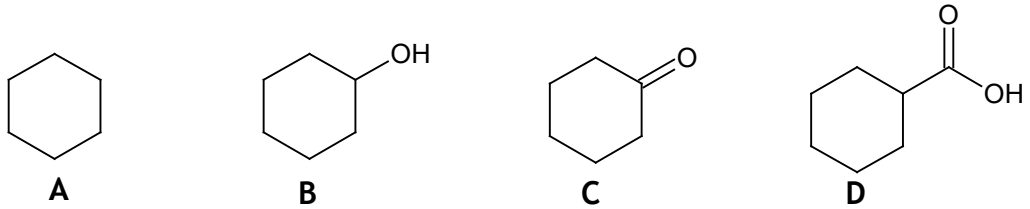
b) Write the semi-structural formula for the following structure. (1 mark)



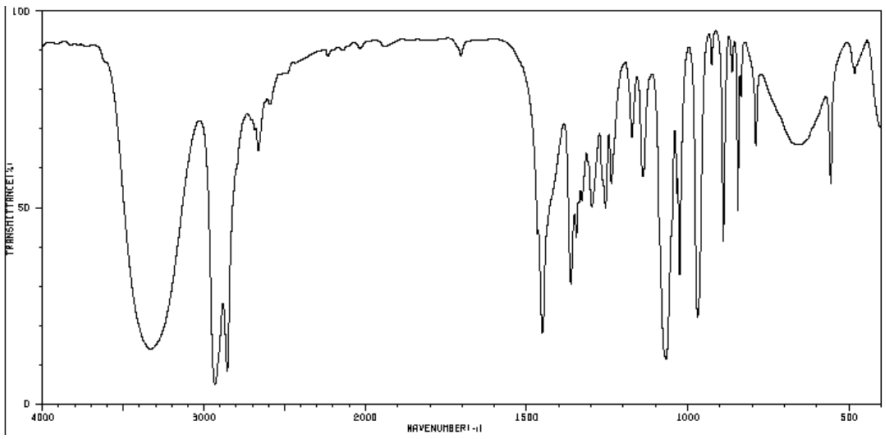
$CH_3CHC(OH)CH_2COOH$ or $CH_3CHCOHCH_2COOH$

Q11. **(3 marks)**

Consider the following four structures.



Which of these is most likely to have produced the following IR spectrum? Carefully justify your choice, by referring to all relevant conclusions that can be drawn from the spectrum. (3 marks)



B (1)

peak at ~3350 cm^{-1} indicative of -OH group (1)

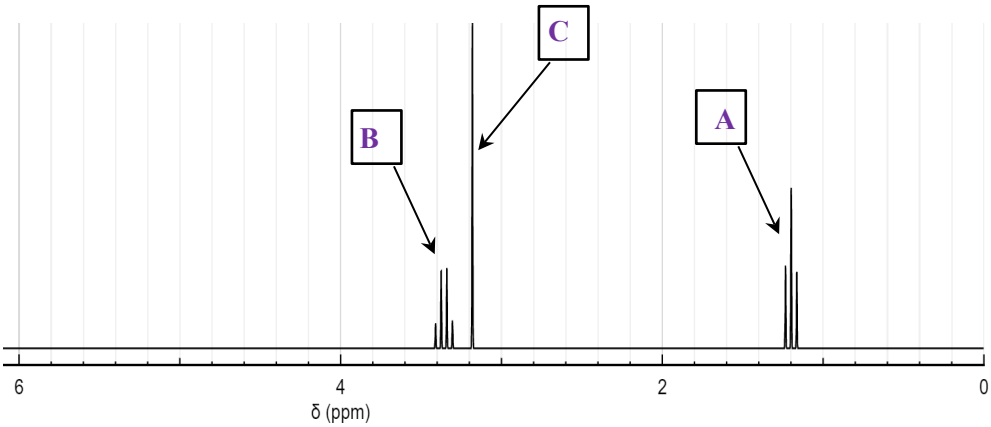
- probably hydroxy

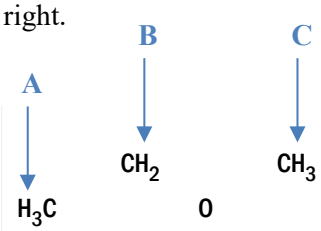
absence of peak at ~1600-1800 cm^{-1} indicates no carbonyl group (C=O) (1)

- no carboxy group and no ketone

Q12. **(6 marks)**

Below is the ^{1}H NMR spectrum of methoxyethane, whose structure is shown at the right.





δ (ppm)	*multiplicity*
1.2	triplet
3.2	singlet
3.4	quartet

a) The table gives details of the three signals recorded. Explain what the number of (low resolution) signals or peaks in the spectrum indicates about the structure. (1 mark)

Number of signals = ***number of H environments***

b) In the boxes provided on the spectrum, clearly identify the hydrogen atoms in the structure (A, B, or C) responsible for each of the signals. (1 mark)

The molecule contains two methyl groups, labelled A and C.

c) Show how the ***chemical shifts in the Data Book*** can be used to distinguish between the signals produced by methyl groups A and C. (2 marks)

Data Book:

R–**CH**$_3$	0.9–1.0
R—O—**CH**$_3$ or R—O—**CH**$_2$R	3.3–3.7

signal at δ = 3.2 is a better match for C, which is adjacent to an O atom

signal at δ = 1.2 is a better match for A, which is adjacent to an -CH_2- group

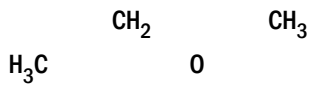
d) Explain how the ***multiplicities*** observed can be used to distinguish between the signals produced by methyl groups A and C. (2 marks)

C, which has no neighboring H atoms, will appear as a ***singlet***

A, which is adjacent to an -CH_2- group and has two neighboring H atoms, will appear as a ***triplet*** (N+1)

Q13. **(7 marks)**

Methoxyethane is mixed with chlorine gas (Cl_2) and irradiated with UV light.



methoxyethane

The dominant reaction can be represented by the following equation:

$$C_3H_8O(g) + Cl_2(g) \xrightarrow{UV\ light} C_3H_7ClO(g) + HCl(g)$$

M(g mol^{-1}):	60.0	71.0	94.5	36.5

a) What ***type*** of reaction is this? (1 mark)

substitution

b) Calculate the atom economy of this reaction. (2 marks)

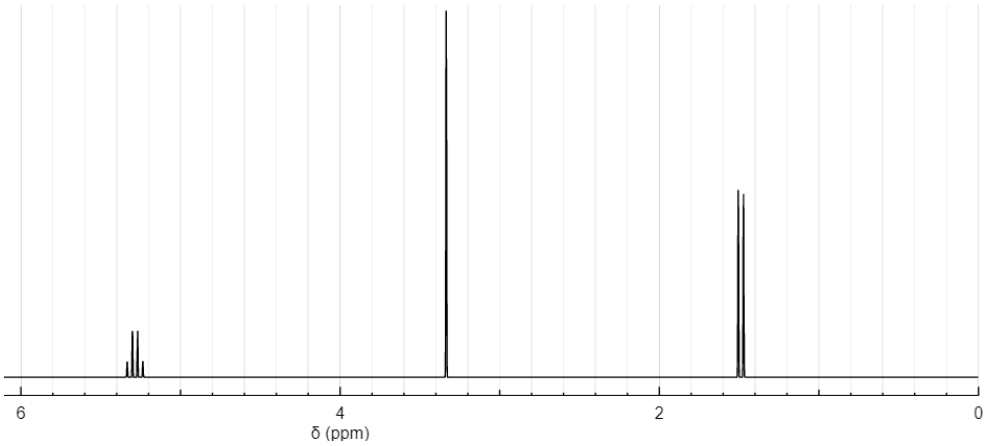
$$atom\ economy = \frac{mass\ of\ target\ product}{total\ mass\ of\ reactants} \times \frac{100}{1} = \frac{94.5}{60.0 + 71.0} \times \frac{100}{1} = 72.1\%$$

c) In one reaction 0.650 g of methoxyethane was reacted, and 0.450 g of the product, C_3H_7ClO, was isolated. Calculate the percentage yield. (2 marks)

$$theoretical\ (100\%)\ yield = \frac{94.5}{60.0} \times 0.650 = 1.024g$$

$$yield = \frac{mass\ obtained}{theoretical\ yield} \times \frac{100}{1} = \frac{0.450}{1.024} \times \frac{100}{1} = 44.0\%$$

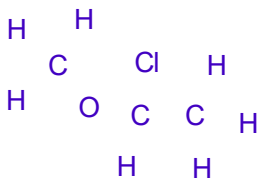
d) The product, C_3H_7ClO, proved to be a mixture of structural isomers, which were separated by distillation. Below is the ^1H NMR spectrum of the predominant isomer obtained.



δ (ppm)	*multiplicity*
1.5	doublet
3.3	singlet
5.3	quartet

Draw the full structure of the product of the reaction that produced this spectrum. (2 marks)

correct structure (1), drawn correctly (all atoms/bonds) (1)

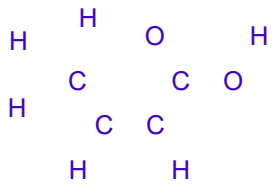


- *most easily determined from the splitting patterns, with the low δ signal having changed from triplet to doublet, indicating the loss of one of the neighboring H atoms*
- *δ values are less useful as the Data Book does not give a predicted range for a H atom adjacent to both O and Cl – their combined deshielding effects account for the signal at 5.3 ppm.*

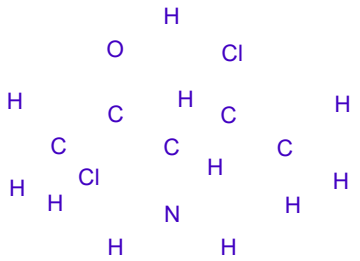
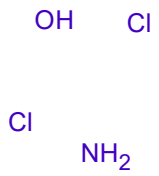
Q14. **(4 marks)**

For each compound named below draw both a complete structure, ***showing all atoms and bonds***, and also a skeletal structure.

a) *cis*-but-2-enoic acid (2 marks)

full structure	skeletal structure
	either

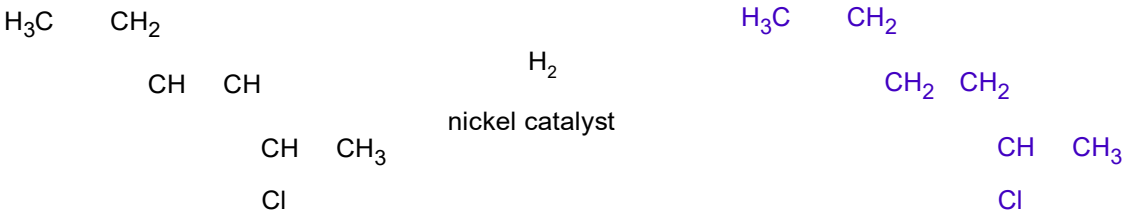
b) 3-amino-2,4-dichloropentan-2-ol (2 marks)

full structure	skeletal structure
	

Q15. **(2 marks)**

In each of the following, draw a structure of the organic product formed (by-products need not be included).

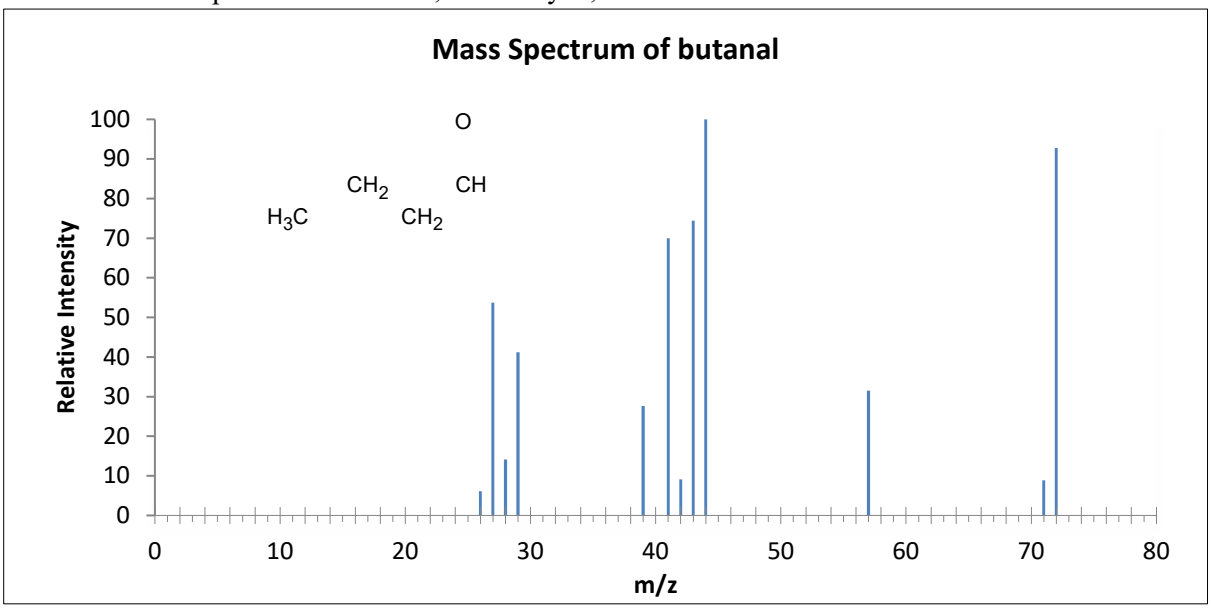
a)



b)

Q16. **(4 marks)**

Below is a mass spectrum of butanal, an aldehyde, whose structure is also shown.



Refer to the spectrum and answer the following questions. (4 marks)

a) What are the m/z values for

i. the parent/molecular ion peak? ....72....

ii. the base peak? ....44....

b) The signals in the spectrum that are listed below can be attributed to fragments arising from simple fission (breaking) of one bond in the parent ion.
Write the formula of a particle formed this way that could be responsible for the signal at

i. m/z = 29 ....$[CH_3CH_2]^+$ or $[CHO]^+$....

ii. m/z = 57 ....$[CH_2CH_2CHO]^+$....

-1 mark if +ve charge omitted

Markers comments – SAC 4 – Organics and Synthesis:

Multiple Choice

- Was very well done although there was a noticeable difference between classes in the Gym compared to the ones in the Science building.
- Question 1
  - Proved difficult as was the most commonly incorrect answer – there was a range of answers given (no common incorrect answer).
- Some struggled with Q3 and 4 as well.

Short Answer

- Question 9

- This was done well on the whole. Nearly everyone got at least 1 mark for both a) and b). There were no half marks given so they must have explained the intermolecular bonding for both molecules to get the other mark.
- There are still boys who make careless mistakes with spelling 'pentate', hexane instead of hexanoic acid. Some do not write clearly and continue to write a paragraph which rambles and doesn't get to the point.

- Question 10 a)
  - There were no part marks. Mostly well done.
- Question 10 b)
  - A few misread and named the molecule rather than giving the semi-structural formula. A few added an extra H.
- Question 11
  - Very well done. There were a few who didn't mention the lack of the peak around 1600-1840cm^{-1} to discount C=O and therefore did not absolutely discount molecule D.
- Question 12 a)
  - Mostly done well, some confused correct order of B,C,A for C,B,A
- Question 12 c)
  - students must quote the <u>type of protons</u> from the data book as
    - Methyl group A = **R-CH$_3$** at ~1.2 ppm (include units)
    - Methyl group C = **R-O-CH$_3$** or R-O-CH$_2$R at 3.2 ppm
- Question 12d)
  - Many students did not mention methyl group A <u>neighbours 2 hydrogen</u> (CH_2) and methyl group C <u>neighbours no hydrogens</u>.
- Question 13 a)
  - Spelling of 'substitution' was not penalised but student should be careful. 'Chlorination' was not accepted as you can have chlorination for an addition reaction
- Question 13b
  - Most achieved full marks here. Students should not round during the answer. Many calculated percentage yield using the theoretical and actual mole quantities, meaning answer of 43.97 %or 44.0% were correct.
- Question 13d)
  - Students should be aware there was only 3-proton environments, hence an alcohol was not possible. Many got the correct structure
- Question 14
  - Many students did not correctly draw the cis isomer in 14a. **Emphasise this**
- Question 16

Many students forgot the <u>positive charge</u> on the fragments.